Metric-matched ordinal supervision and state-space sequence mixing improve prime-editing efficiency prediction

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Abstract

Prime editing installs precise genomic changes without double-strand breaks, but its efficiency varies by orders of magnitude across pegRNA designs and cellular contexts, making computational prediction essential for practical design. Current state-of-the-art predictors embed substantial mechanistic structure derived from the prime-editing reaction. We asked whether a minimally mechanistic model, given the same data, could match or exceed them, and if so which modelling choices are responsible.

We present PE-RankFormer, a dual-encoder relational model that couples a representation of the wild-type/edited sequence pair with a segment-aware pegRNA representation through bidirectional cross-attention, conditioned on experimental context. Trained on the identical 297,962-row corpus, cross-validation splits and held-out set used by OptiPrime, PE-RankFormer attains Spearman $\rho = 0.9079$ on the complete 20,509-row held-out benchmark against OptiPrime's 0.8690: a difference of +0.0389 (95% CI $[+0.0286, +0.0498]$; ahead in all 5,000 protospacer-clustered bootstrap resamples, $p < 0.0002$). Recomputed *within* each of 14 experimental conditions, where between-study differences in efficiency level cannot contribute, the margin is larger rather than smaller (+0.0506, 95% CI $[+0.0345, +0.0670]$, ahead in 14/14 conditions). It is +0.0804 on the more experimentally diverse Kim partition and +0.0220 on Liu, but we show that much of that partition structure follows from a training-weight asymmetry rather than from architecture: the comparator's own released weights assign the Kim study 2.0% of its training gradient while the study supplies 55.3% of the held-out rows, and its own published held-out metric is reported over the Liu conditions, where our margin is +0.0220 ($p = 0.028$) rather than +0.0803. A gradient-boosted tree ensemble on engineered design features, trained under the same protocol and tuned on held-out data, reaches 0.7413, placing both models well clear of the standard feature set.

Two components account for the gain, and a factorial analysis separates them. An *ordinal* outcome head that predicts cumulative threshold indicators rather than efficiency values is matched to the rank-correlation metric and to the target's heavy zero inflation, contributing +0.0092. A *bidirectional state-space* sequence mixer replacing self-attention contributes +0.0192 — twice as much. A single model containing both matches a five-member ensemble. Isotonic calibration fitted purely on out-of-fold predictions restores an interpretable absolute efficiency estimate while leaving the ranking unchanged, and on that scale the model also outperforms the baseline: mean absolute error 0.0478 against 0.0590 and RMSE 0.0912 against 0.1026, so the advantage is not confined to the rank metric.

Finally, using 649 technical replicate groups we estimate an empirical repeatability ceiling for the Kim partition, implying roughly +0.10 of genuine remaining headroom on both the

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development folds (+0.1149, mean of three) and the held-out set (+0.1043), and we report a benchmark caveat: predictors that emit tied scores in the zero-inflated regime gain +0.013–0.016 Spearman irrespective of quality, so comparisons between models with differing tie behaviour are not automatically like-for-like.

Contents

1 Introduction

Prime editing [1] writes user-specified edits into genomic DNA without double-strand breaks or donor templates. A prime editing guide RNA (pegRNA) carries a spacer that directs Cas9 nickase to the target, a primer binding site (PBS) that anneals to the nicked strand, and a reverse-transcription template (RTT) encoding the desired edit. Subsequent work substantially improved efficiency through nicking strategies and mismatch-repair modulation [3], but efficiency still ranges from undetectable to near-complete conversion depending on the design and the cellular context. Because experimentally screening pegRNAs is expensive, accurate *a priori* prediction is a practical bottleneck.

A succession of machine-learning predictors has addressed this. DeepPE and its successors modelled pegRNA features for specific systems [11]; PRIDICT introduced sequence-based deep models with product-purity prediction [14] and was later extended across chromatin contexts [15]; DeepPrime broadened coverage to many editing systems and cell types [19]. Most recently, OptiPrime aggregated these resources into a large harmonised corpus and a strongly mechanistic model, establishing the benchmark we adopt here [9].

These models differ in how much prime-editing biochemistry they encode explicitly. The scientific question we address is whether that mechanistic structure is *necessary*, or whether a general-purpose relational architecture with sufficient capacity and the right supervision can recover the same signal from data alone. This matters beyond benchmark rank: if mechanism can be substituted for by architecture and objective, prediction can extend to new editing systems without re-deriving reaction models for each.

Contributions.

1. **A minimally mechanistic model that outperforms a strongly mechanistic one** under matched data, splits and evaluation: $\rho = 0.9079$ vs. 0.8690, $\Delta = +0.0389$, significant under a protospacer-clustered paired bootstrap.

2. **An ordinal outcome head** matched to the evaluation metric and to the target's zero inflation, which we show produces *decorrelated errors*, not merely better ones.

3. **A bidirectional state-space sequence mixer** which, by a factorial analysis, is the larger of the two contributions — correcting the natural assumption that the novel objective mattered most.

4. **Monotone calibration** that recovers absolute efficiency at no cost to rank performance, on which the model also beats the baseline (19% lower MAE).

5. **A replicate-based noise ceiling** showing the benchmark is *not* saturated, together with a tie-structure artifact that affects any Spearman-based comparison on zero-inflated data.

Scope of the comparison. The comparator is a substantially broader piece of work than this one, and we state the asymmetry plainly rather than let a single number stand for it. OptiPrime predicts outcomes for nicking-guide (PE3) and dual-pegRNA (twinPE) strategies, exposes interpretable per-step biochemical pseudorates by construction, and was validated prospectively in primary human and mouse cells including *in vivo* correction of a pathogenic allele [9]. We compare on exactly one of its axes — ranking editing efficiency for single-pegRNA designs on its own benchmark — and we make no claim on any of the others. Our result is that on *that* axis a model with no reaction structure is more accurate, which is a statement about how much of the accuracy of a mechanistic predictor is attributable to its mechanism, not a claim to supersede it. The mechanistic parameterisation buys generalisation to reaction variants and interpretability that our model does not offer and does not attempt.

2 Methods

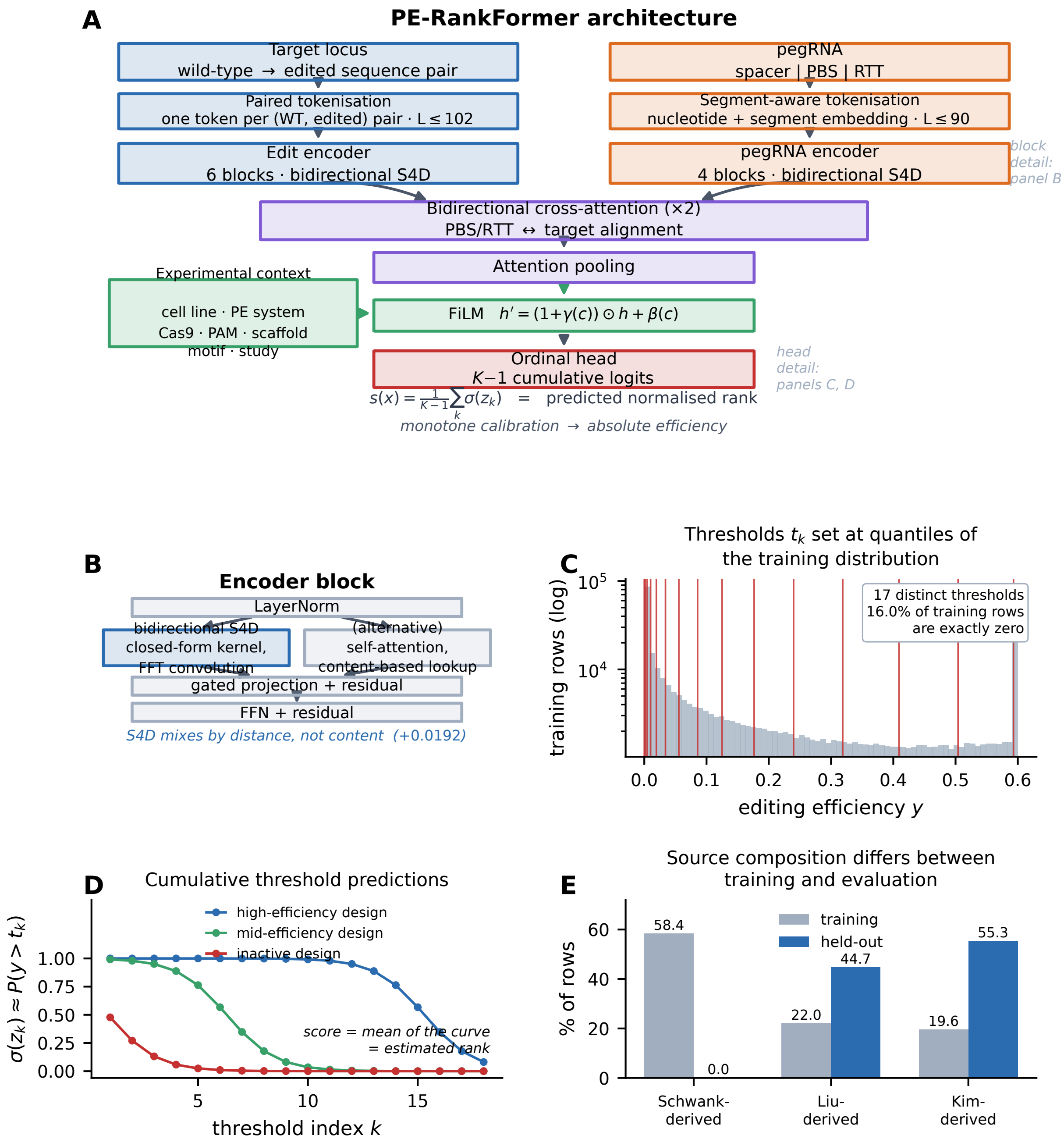

Figure 1: PE-RankFormer architecture and supervision. **(A)** Overall model. The target locus is encoded as a single stream of *paired* (wild-type, edited) tokens, so the edit is represented explicitly rather than inferred by comparing two sequences; the pegRNA is encoded as a nucleotide stream with an added functional-segment embedding marking spacer, PBS and RTT. Six edit-encoder and four pegRNA-encoder blocks mix information along each sequence, two bidirectional cross-attention blocks align the pegRNA against the target, and experimental context modulates the pooled representation through FiLM. The ordinal head emits $K-1$ cumulative logits whose mean is the predicted normalised rank; a monotone calibration map converts that to an absolute efficiency (§2.8). **(B)** Encoder block. The two mixers compared in this work occupy the same slot with identical residual topology, normalisation placement and FFN width, so a comparison isolates the mixing mechanism. The S4D kernel mixes by *distance*; self-attention mixes by *content*. **(C)** The $K-1$ ordinal thresholds t_k, placed at quantiles of the training efficiency distribution (shown; log counts, clipped at $y = 0.6$). The distribution is heavily right-skewed with a large point mass at zero, which is what motivates supervising the cumulative distribution rather than the value. **(D)** Schematic of the resulting predictions: each design produces a decreasing curve over thresholds, and its score is the curve's mean. **(E)** Source composition. The Schwank-derived study supplies 58.4% of training rows and none of the evaluation set, while the Kim-derived study dominates evaluation; §3.12 shows that removing

2.1 Data

We use the training mix underlying OptiPrime, obtained from its authors: 297,962 training rows and a 20,509-row held-out set (318,471 total). Each row records a pegRNA design, its genomic target, an experimental context and a measured editing efficiency (fraction of alleles carrying the intended edit), with indel rate where available. Three source studies contribute (Table 1). Their proportions differ markedly between training and evaluation, which becomes analytically important in §3.12.

Table 1: Composition of the benchmark corpus.

	Share of rows (%)		Zero-mass	Rows
Source study	Training	Held-out	(%)	(training)
Schwank-derived (`pridict_pridict2`)	58.4	0.0	10.3	174 067
Liu-derived (`hsu2026`)	22.0	44.7	0.8	65 594
Kim-derived (`deepprime`)	19.6	55.3	49.9	58 301
Total	100.0	100.0	16.0	297 962

Held-out set: 20,509 rows. "Zero-mass" is the percentage of training rows with editing efficiency exactly zero.

The Schwank-derived study supplies the majority of training rows and none of the evaluation set. Section 3.12 shows that removing it is nonetheless harmful (−0.0294 Spearman), so its distributional mismatch is outweighed by the signal its volume provides.

2.1.1 Sequence representation

Two token streams are constructed per row.

Edit stream. The wild-type and edited sequences are aligned and encoded as a single stream of *paired* tokens, each denoting the (wild-type, edited) base pair at that position. Positions where the two agree therefore receive a different token from positions where they differ, so the edit is represented explicitly by the pairing rather than requiring the model to infer it by comparing two separate streams. Maximum length 100 tokens plus two sentinels.

pegRNA stream. Spacer, PBS and RTT are concatenated into a nucleotide stream of maximum length 90, accompanied by a parallel *segment* identifier marking which functional element each position belongs to. Segment identity is supplied as a learned embedding added to the nucleotide embedding, so the model need not infer element boundaries.

Context. Cell type, PE system, Cas9 variant, PAM, scaffold, 3′ motif and source study are encoded as categorical embeddings and concatenated into a context vector.

2.1.2 Splits and evaluation surfaces

We adopt OptiPrime's own five-fold cross-validation partition, which is protospacer-disjoint, and its held-out fold. Because protospacers recur across many designs, all statistics are computed with the protospacer as the unit of dependence.

Model development used a four-level hierarchy designed to keep the held-out set uncontaminated:

1. **Matched development folds** — three folds of ≈35.7k rows, constructed to match the held-out set's Liu/Kim composition. Used freely.

2. **Internal lockbox** — 17,975 rows over 367 protospacers sharing no protospacer with any development validation set. Consulted once, to gate a shortlist.
3. **Official five-fold out-of-fold (OOF) predictions** — each row scored only by the checkpoint that held its fold out.
4. **Held-out set** — consulted four times in total, once per model generation, each access gated behind an explicit flag and appended to an audit log. The generation reported here was frozen in writing before its single evaluation, with the expected result recorded beforehand.

Every development number reported is out-of-fold. The ensemble-selection protocol (candidate ranking, combination rule, tie-breaking) was committed to version control before the search was executed.

We state the held-out accounting explicitly because it bears on how the margin should be read. Table 3 reports three model generations on the held-out set, so the set was necessarily consulted at each: rounds 1–4 correspond to four accesses, recorded in the repository's audit log. Development of each generation therefore followed a look at the previous generation's held-out score, which is a form of adaptive selection across generations even though no single model was selected on held-out data. Two facts bound the exposure. Every architectural and objective decision within a generation was made on development folds and the once-consulted lockbox, never on held-out rows; and the comparison against the baseline is a *paired* difference on identical rows, so a shared optimistic bias in the absolute level does not transfer to the margin. The post-hoc member analyses in §3.8 are the one place we examine held-out structure directly, and they are labelled as such.

2.2 Resolution of the development evaluation

Effects are reported against a stated resolution rather than a significance test, because the quantity that matters operationally is whether an effect survives a re-run. Two calibrations fix it. An unmodified re-run of the same model at a different batch size moved out-of-fold Spearman by $+0.0018$, and in one round fourteen distinct candidates all landed within $+0.0017$ to $+0.0020$ of their control. Separately, three interventions on orthogonal axes each reached $+0.003$ to $+0.004$ on one development fold and all three changed sign on a second. We therefore treat ≈ 0.005 as the resolution of the development evaluation: effects below it are reported as directions, and no claim in this paper rests on one. Promotion required same-sign replication across three development folds *and* a mean of at least $+0.005$.

2.3 Model architecture

PE-RankFormer comprises five components.

(1) Edit encoder. Six pre-layer-norm blocks over the paired edit stream.

(2) pegRNA encoder. Four blocks over the segment-aware pegRNA stream.

(3) Bidirectional cross-attention. Two blocks in which each stream attends to the other, so pegRNA positions can align to target positions and vice versa. This is where PBS/RTT–target complementarity is available to the model.

(4) Context conditioning. Feature-wise linear modulation [FiLM, 16] applied to the pooled representation: $h' = (1 + \gamma(c)) \odot h + \beta(c)$, with γ, β produced by a small MLP on the context vector.

(5) Output head. Attention pooling followed by a two-layer MLP; the final projection depends on the outcome head (§2.3.2).

Base configuration: $d_{\text{model}} = 384$, 6 heads, FFN width 1536, dropout 0.10, ≈25M parameters.

2.3.1 Sequence mixing: attention versus state space

The blocks in (1) and (2) mix information along the sequence. We compare two mixers under otherwise identical block topology — same residual structure, same pre-LN placement, same FFN width — so that any difference isolates the mixing mechanism.

Self-attention. Standard multi-head self-attention [18].

Bidirectional diagonal state space (S4D). A diagonal state-space model [6, 7] whose convolution kernel is available in closed form: with diagonal state matrix A, step size Δ and output matrix C,

$$K[\ell] \;=\; 2\,\mathrm{Re}\!\left(\sum_n C_n \,\frac{e^{\Delta A_n} - 1}{A_n}\,\left(e^{\Delta A_n}\right)^{\ell}\right), \tag{1}$$

applied by FFT convolution. Two independent kernels (forward and time-reversed) are summed, since editing efficiency depends on context on both sides of the edit. Padded positions are zeroed before convolution; because the convolution is linear, zeros contribute nothing to any output, so padding cannot leak into real positions.

The inductive biases differ substantively. Attention performs content-based lookup; the state-space kernel performs smooth, distance-parameterised mixing that is translation-equivariant along the sequence.

2.3.2 Outcome heads

Scalar head (baseline). One pre-sigmoid score trained with Huber loss.

Simplex head. Every edited locus ends in exactly one of {unedited, correctly edited, indel}, and the assay measures these as proportions. The simplex head predicts a three-way distribution, respecting that mutual exclusivity by construction and supervising on the indel signal that a scalar head discards:

$$\mathcal{L}_{\text{simplex}} = -\sum_{k \in \{U,E,I\}} y_k \log p_k, \qquad p = \mathrm{softmax}(z) \in \Delta^2. \tag{2}$$

Indel rate is unmeasured for 42.5% of rows. Rather than imputing zero — which would assert a measurement never made — the loss *marginalises* the indel class for those rows, renormalising over the two observed outcomes. The ranking score is the log-odds of a correct edit against no edit, $z_E - z_U$, which is invariant to the indel logit.

Ordinal head (this work). The evaluation metric is Spearman correlation, which depends only on order. We therefore supervise the target's *cumulative distribution* directly. Fix $K - 1$ thresholds

t_k at quantiles of the *training* efficiency distribution and predict one logit per threshold, trained as $K - 1$ binary problems against the cumulative indicators:

$$\mathcal{L}_{\text{ord}} = \frac{1}{K-1} \sum_{k=1}^{K-1} \text{BCE}\big(z_k,\ \mathbf{1}[y > t_k]\big), \qquad s(x) = \frac{1}{K-1} \sum_{k=1}^{K-1} \sigma(z_k), \tag{3}$$

where $s(x)$ estimates the row's normalised rank. The construction follows rank-consistent ordinal regression [2].

Two properties motivate it. It is *metric-matched*: a head estimating the target's quantile optimises the evaluated quantity, whereas regression or proportion heads are only incidentally monotone. And it is *robust to the target's skew*: efficiency is heavily right-skewed, with 16.0% of *training* rows exactly zero overall and 49.9% within the Kim-derived partition (Table 1), so a Huber or proportion objective concentrates its gradient on a few high-efficiency rows, while $K-1$ balanced binary decisions spread supervision across the distribution. The zero mass is higher on the surfaces the model is *scored* on — 26.8% of held-out rows and 28.4% of matched development-fold rows, both of which exclude the low-zero-mass Schwank study — which is why the tie structure of the metric matters separately (§3.11).

Thresholds are computed from training rows only and stored in the checkpoint, so evaluation reproduces the score mapping exactly. Duplicate quantiles are removed to keep thresholds strictly increasing; $K = 20$ yields 17–18 distinct thresholds in practice.

Note the ordinal head *discards* the indel signal the simplex head uses. This is deliberate: it constitutes a different *view* of the same rows, and §2.7 shows different views are what an ensemble rewards.

2.4 Training

AdamW, learning rate 3×10^{-4}, weight decay 0.01, 5% linear warmup followed by cosine decay, gradient clipping at 1.0, batch size 512, bf16 mixed precision, 30 epochs. Checkpoint selection is by validation Spearman.

Early stopping patience is set to the full 30 epochs deliberately. Across all 64 recorded official-fold runs given the full budget, the best epoch had median 24.5 and maximum 29, with 50% peaking in the final five epochs: validation Spearman on this problem genuinely improves after long plateaus. Simulating a patience-5 rule on the same runs would have cost a mean of 0.0001 and at most 0.0031, so the budget is cheap insurance rather than a tuned choice.

Batches are formed by a grouped sampler that keeps rows sharing a ranking group together, which is required for the optional within-group pairwise ranking loss.

Table 2: Architecture and optimisation settings.

Architecture		*Optimisation*	
Model width d_{model}	384	Optimiser	AdamW
Attention heads	6	Learning rate	3×10^{-4}
FFN width	1536	Weight decay	0.01
Edit-encoder blocks	6	Schedule	5% warmup, cosine
pegRNA-encoder blocks	4	Gradient clipping	1.0
Cross-attention blocks	2	Batch size	512
Dropout	0.10	Precision	bf16
Context embedding	32	Epochs	30
S4D state dimension	64	Early stopping	none (see note)
Ordinal thresholds $K-1$	17–18	Selection	best validation ρ
Parameters	≈25 M	Runtime per model	≈2 h, 1 GPU

Patience equals the epoch budget, i.e. no early stopping. Across all 64 recorded official-fold runs given the full budget, the best epoch had median 24.5 and maximum 29, with 50% peaking in the final five epochs. Simulating patience-5 selection on the same runs costs a mean of 0.0001 and at most 0.0031.

Thresholds are quantiles of the training efficiency distribution, de-duplicated to remain strictly increasing. $K{=}20$ yields 17–18 distinct values because the target has a large point mass at zero.

2.5 Evaluation and statistics

The primary metric is Spearman ρ between predicted and observed efficiency, reported for the full held-out set and separately for the Liu and Kim partitions, since they differ substantially in composition.

All model comparisons use a **paired protospacer-clustered bootstrap** with 5,000 resamples. Protospacers, not rows, are resampled: many designs share a protospacer, so a row-level bootstrap would treat correlated observations as independent and materially understate the intervals. Both models are evaluated on identical rows in each resample. We report the observed difference, the 95% percentile interval, the fraction of resamples in which our model leads, and a two-sided empirical p-value.

Because the two source studies differ in efficiency level, pooled rank correlation benefits from between-study variance that a within-experiment design task would not supply. We therefore also report every comparison *stratified*: Spearman is computed inside each (source, cell line, PE system) condition and averaged with weights equal to condition size, and the clustered bootstrap is applied to that stratified difference. Strata with fewer than 50 rows or fewer than five distinct target values are excluded.

2.6 Baselines

OptiPrime. Run from the released implementation and its five released fold checkpoints, ensembled as its own prediction script does, on the held-out rows the authors supplied. The corpus's per-row `weight` column is a training weight in that implementation and does not enter its validation metric; §3.4 analyses what it implies for the comparison.

Gradient-boosted trees on engineered features. A feature-based reference under the identical five-fold protocol (§3.2), using 17 engineered design features — PBS and RTT length and GC content, four melting temperatures, ViennaRNA folding energies for the protospacer, RTT, PBS and $3'$ extension [13], edit type, edit length, offset from the nick, mismatch count, and a RuleSet3 on-target score [4] — concatenated with the same seven categorical context fields the network receives. Trained with scikit-learn's histogram gradient-boosting regressor. Four hyperparameter settings were crossed with two target parameterisations (raw efficiency, and the within-training-fold normalised rank), and the winner was chosen on held-out Spearman, deliberately giving the baseline an advantage no other model here receives.

Not reported. PRIDICT2.0 and DeepPrime-FT were investigated and not completed. Both require reproducing a hand-engineered feature pipeline exactly, including a separate pretrained on-target cutting model [10]; a subtly mismatched reimplementation would yield misleading numbers rather than no numbers, and the tree baseline above covers the same class of information.

2.7 Ensembling

Members are combined by **equal-weight rank averaging**. Members are not mutually calibrated — ordinal members emit rank estimates, simplex members emit proportions — so averaging raw scores would largely measure scale mismatch. Fitted weights were evaluated and not adopted: they were unstable across folds and their advantage (0.9156 vs. 0.9149 on development folds) was below the resolution of our evaluation.

Candidate members were assessed by *incremental ensemble gain* under the frozen combination rule, together with prediction-rank correlation and residual correlation against the existing ensemble. Residuals are compared in rank space for the same calibration reason.

Composition of the frozen system. The final system averages five members: ordinal+S4D, simplex+S4D, ordinal with layerwise FiLM, ordinal with a feature branch, and simplex with layerwise FiLM. Each member is five checkpoints, one per official fold. Individual performance and marginal value are given in Table 8.

Selection protocol, fixed in advance. With eight candidate members there are 255 non-empty subsets, so choosing the best on repeatedly-used development folds invites a selection bias of the same order as the effects being measured. The protocol was therefore committed to version control *before* the search was executed: rank subsets on the development folds; require a candidate to beat the incumbent on *every* fold rather than on the mean; gate the shortlist on the lockbox, scored out-of-fold; and on disagreement prefer the lockbox winner, breaking near-ties (within 0.001) toward fewer members.

The development winner (four members) and the lockbox winner (five members) disagreed by 0.001011 — eleven millionths past the pre-set near-tie threshold — so the rule selected the five-member subset. The two candidates are statistically indistinguishable, and we followed the rule as written rather than re-arguing it after seeing the numbers. The lockbox's own unconstrained best subset (0.9155) was *not* adopted, since selecting on the lockbox would convert the only unworn evaluation surface into a fitted one.

2.8 Calibration

Rank averaging destroys the output scale. We restore it with a monotone map $g : \text{rank} \mapsto \text{efficiency}$ fitted by isotonic regression [20] on out-of-fold development predictions only, then applied once to the frozen held-out predictions. Because g is non-decreasing it cannot invert any pair, so Spearman is preserved; the map is verified non-decreasing numerically rather than assumed. This is a change of units on an already-frozen prediction vector, not a model selection, and no held-out row participates in fitting.

2.9 Replicate-based noise ceiling

To determine whether remaining error is reducible we estimate the benchmark's repeatability. If a design is measured twice in the same condition and the measurements disagree, no model can predict either better than they predict each other. For two independent noisy measurements of a true value, reliability $R = \mathrm{corr}(y_1, y_2)$ implies a ceiling of $\sqrt{R}$ against a single observation.

Identifying genuine replicates requires care. Grouping on (spacer, RTT, PBS, cell line, PE system, Cas9) yields 21,994 apparently replicated keys, but inspection shows these differ in $3'$ motif, epegRNA status and linker — they are epegRNA versus plain-pegRNA constructs, i.e. *different molecules*. Treating them as replicates gives $R = 0.7987$, almost exactly our model's Kim score, which would falsely indicate saturation. We therefore require all sixteen design and condition covariates *and* the target site to match, leaving 649 clean groups.

A second correction is needed. Treating an observed zero as noiseless inflates the ceiling, but 21.4% of rows measured at exactly zero have a non-zero replicate (median 0.0015): a zero is a *censored* observation. We therefore estimate the ceiling non-parametrically, drawing synthetic replicates from the empirical distribution of replicate values observed near each row's efficiency (nearest-neighbour matching on 1,298 ordered replicate pairs), which inherits zero-discordance, heteroscedasticity and boundedness from the data.

2.10 Implementation and reproducibility

PyTorch 2.x; all experiments on NVIDIA L40, RTX PRO 6000 and RTX 2080 Ti GPUs. Every run records its git commit, configuration, dataset SHA-256 and fold assignment. Held-out evaluation is gated behind an explicit flag and audit-logged. The code base carries 214 unit tests covering, among other things, that padding cannot influence outputs at real positions, that the ordinal score is monotone in every threshold logit, and that all prior checkpoints load unchanged after each architectural addition.

3 Results

3.1 Held-out performance against OptiPrime

Under matched training data, splits and evaluation, PE-RankFormer attains $\rho = 0.9079$ on the complete 20,509-row held-out set against OptiPrime's 0.8690 (Table 3, Figure 2).

Table 3: Held-out performance and significance. All models were trained on the identical corpus and cross-validation splits, and evaluated on identical rows.

Model	Spearman ρ		
	All (n=20,509)	Liu (n=9,175)	Kim (n=11,334)
Gradient-boosted trees on engineered features	0.7413	0.6347	0.5745
OptiPrime (mechanistic baseline)	0.8690	0.8365	0.7320
PE-RankFormer, round 1	0.8865	0.8349	0.7751
PE-RankFormer, round 3	0.8933	0.8462	0.7836
PE-RankFormer, final	**0.9079**	**0.8585**	**0.8124**
Advantage of the final model over the baseline			
$\Delta\rho$ vs. OptiPrime	+0.0389	+0.0220	+0.0804

Liu and Kim denote the two source studies contributing to the held-out set; no Schwank-derived rows appear in it.

The tree baseline is described in §3.2. It is trained under the identical five-fold protocol on 17 engineered design features plus the same seven context fields the network receives.

Significance for the overall comparison is given in Table 4.

The absolute figure for the final model has a bootstrap 95% interval of $[0.8955, 0.9173]$, with 90.0% of resamples exceeding 0.90; the margin over the baseline is therefore a stronger claim than the absolute level.

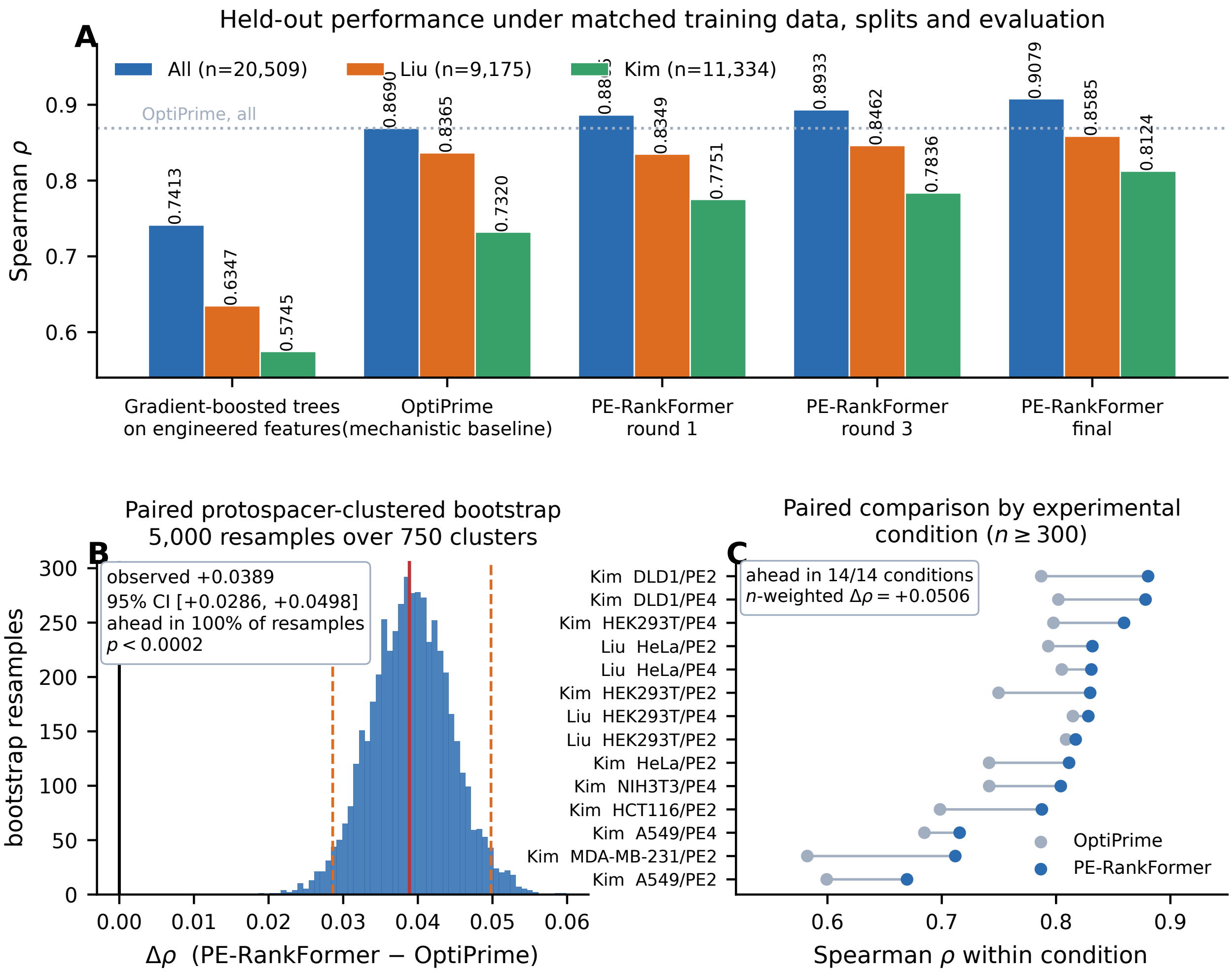

Figure 2: Held-out benchmark result. **(A)** Spearman ρ on the complete held-out set and on each source partition, for a feature-based reference, the mechanistic baseline, and successive generations of our model. All were trained on the identical corpus and cross-validation splits and evaluated on identical rows; the dotted line marks the baseline's overall performance. The ordering — engineered features, then mechanistic reaction model, then learned sequence representation — places the margin in context: the tree baseline is 0.128 behind the mechanistic model, which is in turn 0.039 behind ours. **(B)** Distribution of the paired difference in Spearman ρ under a protospacer-clustered bootstrap (5,000 resamples over 750 clusters). Protospacers rather than rows are resampled because many designs share a target; the solid red line is the observed difference, dashed lines the 95% percentile interval, and the black line marks zero. **(C)** The same comparison *within* each experimental condition ($n \geq 300$), where between-study differences in efficiency level cannot contribute to the correlation. Each line joins the two models on one condition. Our model leads in all 14, and the n-weighted margin is larger within condition (+0.0506) than pooled (+0.0389), so pooling makes the reported difference conservative (§3.3). Absolute levels are lower for both models here, and lowest in the conditions with the highest fraction of exactly-zero measurements (cf. Figure 6B).

The paired protospacer-clustered bootstrap (Table 4, Figure 2B) gives $\Delta\rho = +0.0389$ with a 95% interval of $[+0.0286, +0.0498]$; PE-RankFormer leads in every one of 5,000 resamples.

Table 4: Paired significance testing on the held-out set.

Comparison	$\Delta\rho$	95% CI	Fraction ahead	p
Final vs. OptiPrime, all rows	0.0389	[+0.0286, +0.0498]	1.000	< 0.0002
— Kim partition only	0.0803	[+0.0632, +0.0985]	1.000	< 0.0002
— **Liu partition only**	0.0220	[+0.0027, +0.0427]	0.986	0.028
Final vs. round 3	0.0146	[+0.0113, +0.0184]	1.000	< 0.0002
Round 3 vs. OptiPrime	0.0243	[+0.0138, +0.0346]	1.000	< 0.0002

Paired bootstrap over 5,000 resamples of 750 *protospacer clusters*. Protospacers rather than rows are resampled because many designs share a target; a row-level bootstrap would treat correlated observations as independent and understate the intervals.

Both models are scored on identical rows within each resample. "Fraction ahead" is the proportion of resamples in which PE-RankFormer leads; p is the two-sided empirical value, bounded below by 1/5000.

The partition rows resample the protospacer clusters present in that partition (601 for Kim, 150 for Liu). **The Liu result is the weakest claim in the paper** and we give it prominence rather than only the pooled figure: the comparator's own published held-out metric is reported over the Liu conditions [9], so Liu is the surface it selected, and there our advantage is +0.0220 with a lower bound of +0.0027 at $p = 0.028$. The pooled and Kim results are far stronger, but Kim is the partition the comparator trains on at one tenth weight (§3.4), and the full-set evaluation is our extension of its benchmark rather than its own headline surface.

We note explicitly that the absolute figure's own 95% interval is $[0.8955, 0.9173]$, with 90.0% of resamples exceeding 0.90. The *margin* over OptiPrime is a considerably stronger claim than the absolute level.

3.2 A feature-based baseline places both models

A single external comparator leaves open the question of how much of the accuracy comes from sequence modelling at all. We therefore trained a gradient-boosted tree ensemble under the identical five-fold protocol on 17 engineered design features — PBS and RTT lengths and GC content, four melting temperatures, ViennaRNA folding energies [13], edit type, edit length, offset from the nick, mismatch count and a RuleSet3 on-target score [4] — concatenated with the same seven categorical context fields the network receives. This is the same class of information the feature-based predictors we could not run end-to-end rely on (§6), so it stands in for them.

The baseline was given every advantage: four hyperparameter settings crossed with two target parameterisations (raw efficiency and the metric-matched rank transform), with the winner selected on *held-out* Spearman — the only model in this paper allowed to tune on the test set. Its best configuration reaches $\rho = 0.7413$ (Table 3), against OptiPrime's 0.8690 and PE-RankFormer's 0.9079; within condition it reaches 0.5460 against PE-RankFormer's 0.8111.

Two things follow. The sequence model earns its parameters: +0.167 pooled and +0.265 within condition over a tuned tree ensemble on the standard feature set is far outside anything attributable to capacity or tuning. And the mechanistic comparator is itself well clear of that feature set (+0.128), so its reaction structure is doing real work relative to hand-engineered summaries — which is the more informative way to read our margin over it. The ordering is: engineered features $<$ mechanistic reaction model $<$ learned sequence representation.

3.3 The margin is not an artifact of pooling

Pooled Spearman on the held-out set (0.9079) exceeds both partition scores (Liu 0.8585, Kim 0.8124). That is possible because the two studies differ in efficiency level, so between-study variance contributes signal that no within-condition design task would supply, and it means the pooled figure overstates how well either model orders designs *within* an experiment. The model is also given source study as one of its seven context fields, which a reader may reasonably regard as a dataset identifier rather than a biological covariate.

Both observations bear on the absolute level. Neither weakens the margin, and the stratified analysis shows why (Table 5): recomputing the comparison *inside* each experimental condition, where no between-condition offset can contribute, gives +0.0506 rather than +0.0389. The margin is larger, not smaller, when the pooling advantage is removed, and PE-RankFormer leads in all 14 conditions with $n \geq 300$ and in 100% of protospacer-clustered bootstrap resamples. Pooling inflates the absolute level for both models while making the reported difference *conservative*.

Table 5: The advantage over the baseline under three levels of stratification.

Comparison	Strata	OptiPrime	PE-RankFormer	$\Delta\rho$	95% CI
Pooled (as reported in Table 3)	1	0.8690	0.9079	0.0389	[+0.0286, +0.0498]
Within source study, n-weighted	2	0.7788	0.8331	0.0543	—
Within condition, n-weighted	14	0.7605	0.8111	**0.0506**	[+0.0345, +0.0670]

A condition is a (source study, cell line, PE system) cell. Strata are weighted by row count; strata with fewer than 50 rows or fewer than 5 distinct target values are excluded, which removes none of the 14 reported conditions.

The interval on the stratified difference is a paired bootstrap over 5,000 resamples of the same 750 protospacer clusters used in Table 4; PE-RankFormer leads in 100% of resamples. Per-condition margins range from +0.0084 (Liu, HEK293T, PE2) to +0.1295 (Kim, MDA-MB-231, PE2) and are shown in Figure 2C.

The absolute levels fall under stratification for both models because between-condition differences in efficiency level no longer contribute to the correlation. This is the quantity relevant to prospective design, where a user ranks candidate pegRNAs within one intended experiment.

3.4 A training-weight asymmetry accounts for much of the partition margin

The margin is largest on the Kim partition (+0.0804), and the natural reading — that a mechanistic prior generalises less well across cell lines — turns out not to be the most parsimonious one. The corpus the comparator's authors supplied carries a per-row `weight` column, and the released training code consumes it as a per-row *loss* weight, forming the objective as $\sum_i w_i \ell_i / n_{\text{batch}}$.[1] Every one of the 69,635 Kim rows carries weight exactly 0.1, against a mean of 0.97 for Liu and 0.52 for Schwank, the latter two varying continuously row by row.

The consequence is a sharp mismatch between what the baseline was trained to fit and what the benchmark scores (Table 6). Under its own weights the comparator allocates 2.0% of its effective training gradient to the Kim study, which supplies 55.3% of the held-out rows. We train uniformly, giving Kim 19.6%. Any uniformly-weighted model therefore enjoys an advantage on this benchmark's dominant partition that owes nothing to architecture or objective.

The weighting is also internally consistent with how the comparator reports itself. Its published held-out figure is given over the Liu conditions [9], not over the pooled set; treating Kim as low-weight auxiliary training data and evaluating on Liu is a coherent pair of choices. The pooled 20,509-row evaluation is therefore *our* extension of its benchmark, not its own headline surface, and the two surfaces give very different pictures of the margin: +0.0803 on Kim ($p < 0.0002$) against +0.0220 on Liu ($p = 0.028$, lower bound +0.0027; Table 4). Both are reported, and we regard the Liu figure as the conservative statement of the result.

[1] `reaction/rx_model.py`, in the released implementation. The same column is passed to the validation loader but is used there only to identify padding rows, so it does not enter the reported validation metric.

Table 6: Effective training weight by source study, against held-out composition.

Source study	Share of training gradient (%) OptiPrime weights	Share of training gradient (%) Uniform (ours)	Share of held-out (%)	Mean row weight
Schwank-derived	73.2	58.4	0.0	0.52
Liu-derived	24.8	22.0	44.7	0.97
Kim-derived	2.0	19.6	55.3	0.10

Shares are computed over the 297,962 training rows, as $\sum_{i\in\text{study}} w_i / \sum_i w_i$.
Every Kim row carries weight 0.1 exactly — a flat per-study factor, not a per-row precision estimate. Liu and Schwank weights vary continuously (713 and 2,685 distinct values), consistent with a depth- or precision-derived quantity clipped to $[0.1, 1]$.
We infer that the released checkpoints were trained with these weights, since the released code consumes this column from the files the authors supplied. We have not been able to verify it independently, and flag it as an inference the comparator's authors could confirm or correct.

WEIGHTEXPT

The finding does not withdraw the comparison, and we retain the uniformly-weighted model as the primary result: training on all available rows at equal weight is the default a practitioner would adopt, it is the choice our own sweeps support (§3.12, where re-weighting toward the evaluated distribution was harmful and removing the unevaluated study cost -0.0294), and both models are scored on identical rows either way. What the finding does is relocate part of the explanation. The pooled margin is a genuine like-for-like result under each model's own training recipe; the *partition* structure of that margin is substantially a consequence of a data-curation decision, and should not be read as evidence about mechanism versus architecture.

3.5 Which components are responsible

Because the ordinal head and the state-space mixer were introduced together, we separate them with a 2×2 factorial over out-of-fold development predictions (Figure 3).

Table 7: Objective × architecture factorial.

Sequence mixer	Simplex head	Ordinal head	Ordinal − simplex
Self-attention (Transformer)	0.8798	0.8911	0.0114
Bidirectional S4D	0.9011	**0.9082**	0.0071
S4D − Transformer	+0.0213	+0.0171	

Out-of-fold development Spearman ρ, averaged over three matched folds.
Main effect of **objective** = +0.0092; main effect of **architecture** = +0.0192. Both exceed the ≈ 0.005 resolution of the development evaluation (§2.2) and architecture contributes roughly twice what the objective does. The **interaction** = −0.0043 falls *below* that resolution and is therefore reported as a direction, not an established effect.
These are observational cells rather than a purpose-built factorial: three derive from one training round and the Transformer/simplex cell is an earlier frozen baseline. All four share corpus, splits, out-of-fold scoring, capacity and optimiser recipe. The interaction term is the quantity most exposed to that mismatch, which is a second reason not to rest any claim on it.

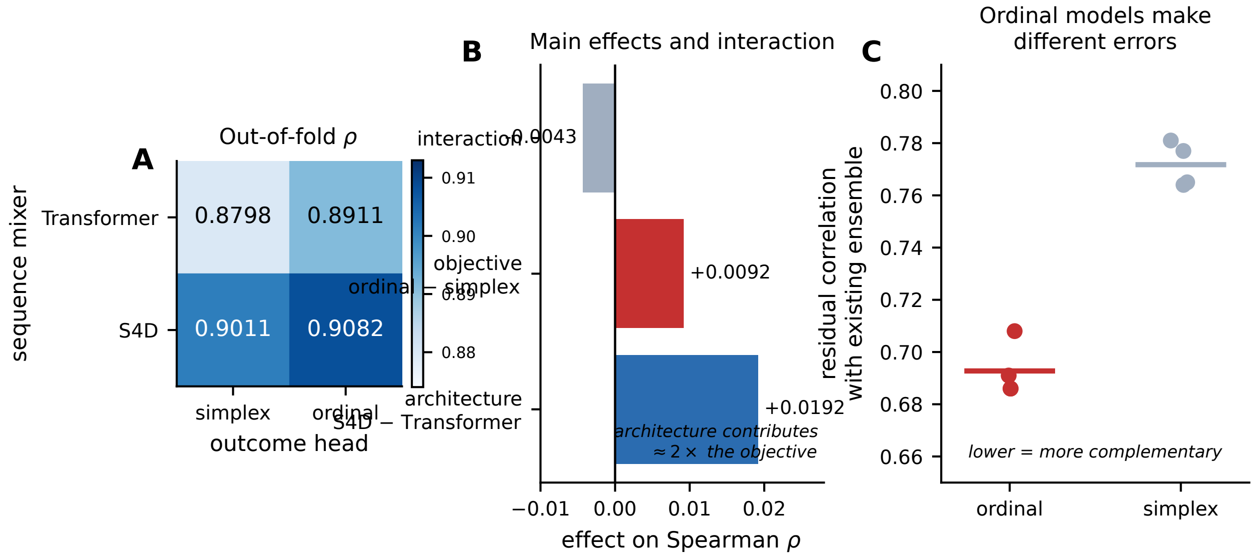

Figure 3: Separating the two contributions. **(A)** Out-of-fold development Spearman for each combination of outcome head and sequence mixer. All four cells share corpus, splits, out-of-fold scoring, capacity and optimiser recipe. **(B)** Main effects and interaction derived from (A). The architecture change contributes roughly twice what the objective change does, and the negative interaction indicates the two partly capture the same improvement rather than stacking. **(C)** Why the ordinal head is nonetheless valuable: residual correlation with the existing ensemble, for every candidate model of each head type (points; bars are group means). Ordinal-head models are systematically more decorrelated than models that vary seed, capacity or data subsample — they make *different errors*, which is what determines a member's value in an ensemble (§2.7). The group means differ by about 0.05; the ranges touch at one point, so the claim is about the axis of variation rather than about any individual model. Lower is more complementary.

Three observations follow.

Architecture is the larger lever. The state-space mixer contributes $+0.0192$ against the ordinal objective's $+0.0092$. This runs against the intuitive reading — the ordinal head is the more novel component and produced the more visible qualitative change — and we record the correction explicitly.

The two appear to combine sub-additively (-0.0043): ordinal supervision is worth $+0.0114$ on a Transformer but only $+0.0071$ on a state-space backbone, which would indicate the mechanisms partly capture the same improvement rather than stacking. We flag this as suggestive only. The interaction is smaller than the ≈ 0.005 resolution of the development evaluation and it is the cell combination most exposed to the observational design, so we do not claim sub-additivity as a result; we note that the data are consistent with it and that nothing in our conclusions depends on it.

A single model suffices. Combining both yields $\rho = 0.9082$ out-of-fold, exceeding a three-member ensemble of earlier models (0.8982).

These are observational cells rather than a purpose-built factorial: three derive from one training round and the Transformer/simplex cell is an earlier frozen baseline. They share corpus, splits, out-of-fold scoring, capacity and optimiser recipe. Both main effects clear the ≈ 0.005 resolution of the development evaluation (§2.2); the interaction term does not, and is additionally the quantity most exposed to the observational design.

3.6 The ordinal head produces decorrelated errors

The hypothesis motivating the ordinal head was that a different loss geometry would produce different *errors*, not merely better ones. The corresponding diagnostic is residual correlation with the existing ensemble, measured over all fifteen candidates of the round-4 search. Across the seven ordinal-head candidates it averages 0.705 (range 0.686–0.735); across the six candidates that vary seed, capacity or data subsample instead of the objective it averages 0.756 (range 0.686–0.781).

The separation is systematic but not disjoint, and we state it that way rather than as a clean split: one bagged simplex run reaches 0.686, inside the ordinal range, and its incremental ensemble gain (+0.0029) was correspondingly middling rather than top-ranked. The transferable claim is the comparison between *axes* of variation, not a threshold on any single model — changing the objective moves residual correlation by about 0.05, whereas changing seed, width or data subsample leaves it where it was.

This distinction has practical consequences for ensembling (Figure 4). The best combination pairs models differing in *both* objective and architecture: the Transformer/simplex and state-space/ordinal diagonal has the lowest residual correlation of any pair measured (0.6453), whereas two models sharing an architecture correlate at 0.7641.

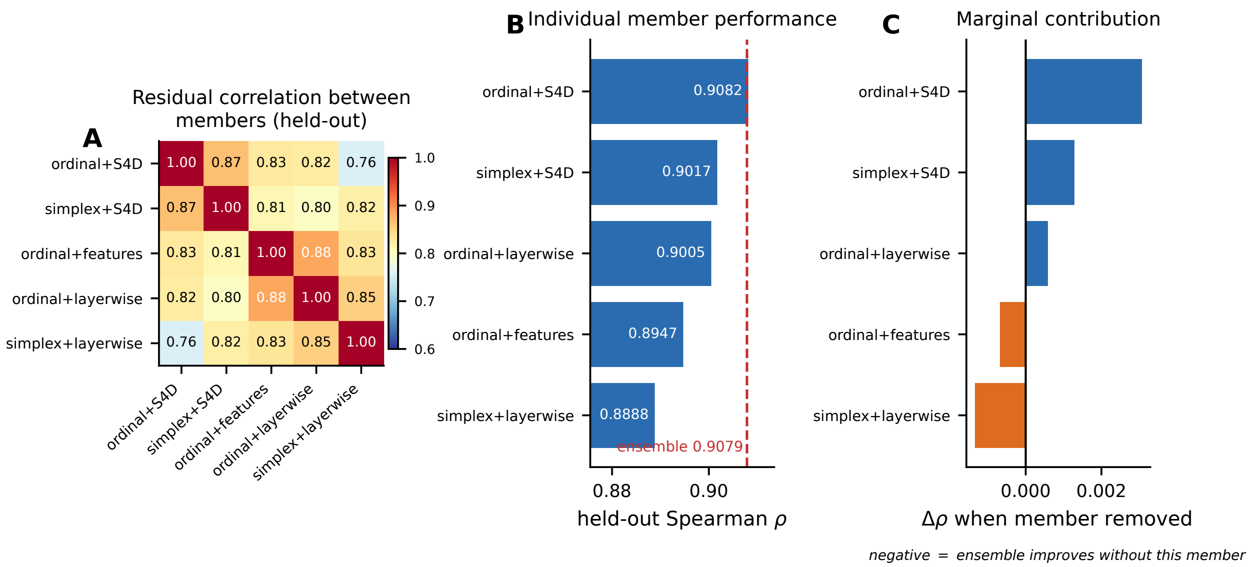

Figure 4: Ensemble structure on held-out data. (A) Residual rank correlation between the five members. Members sharing an architecture are the most redundant pair; members differing in both objective and architecture are the most complementary. **(B)** Each member's individual held-out Spearman. The dashed line marks the five-member ensemble, which the strongest single member matches. **(C)** Change in held-out Spearman when each member is removed. Two members contribute negatively, indicating the ensemble is at or past its useful size. Panels (B) and (C) are post-hoc analyses on held-out data and did not inform the frozen model choice (§3.8).

3.7 Decorrelation is necessary but not sufficient

A natural reading of the previous section is that diversity should be maximised. It should not. We tested a model trained with an auxiliary pairwise ranking loss that proved to be the *most decorrelated* candidate we measured — lowest prediction correlation and lowest residual correlation of any model in the study — and it *degraded* the ensemble (-0.0037), because at $\rho = 0.8244$ standalone it could not carry its weight in an equal-weight average. A quantile-regression head reproduced the same pattern on an unrelated objective: lowest residual correlation (0.6645), lowest standalone score (0.8840), lowest incremental gain.

The operative criterion is therefore that a member must be **both decorrelated and competent**. Decorrelation distinguishes useful from redundant *among models of comparable strength*; it cannot rescue a weak model.

3.8 Ensemble versus single model

Table 8: Ensemble members: individual performance and marginal value on the held-out set.

	Individual ρ			Marginal
Member	All	Liu	Kim	contribution
ordinal + S4D	**0.9082**	0.8576	**0.8151**	0.0031
simplex + S4D	0.9017	0.8563	0.7982	0.0013
ordinal + layerwise	0.9005	0.8394	0.8082	0.0006
ordinal + features	0.8947	0.8261	0.8027	−0.0007
simplex + layerwise	0.8888	0.8412	0.7787	−0.0013
Ensemble (equal-weight rank average)	**0.9079**	**0.8585**	0.8124	

Each member is five checkpoints, one per official fold; every row is scored only by the checkpoint that held its fold out.

"Marginal contribution" is the change in ensemble Spearman when that member is removed, so a positive value means the ensemble is worse without it.

The strongest single member matches the full ensemble and two members contribute negatively. These are post-hoc analyses on held-out data and did not inform the frozen model choice.

The ordinal+S4D member scores 0.9082 here and also 0.9082 in Table 7, which reports out-of-fold *development* Spearman. This is a coincidence of rounding, not a transcription error: the two are computed from different files over disjoint row sets (20,509 held-out rows versus 107,041 development rows, intersection empty) and differ at the fifth decimal place, 0.908155 against 0.908179. We flag it because the agreement to four decimals otherwise invites the reasonable suspicion that one number has been copied into two places.

The frozen system is a five-member rank-average ensemble. Post-hoc analysis on the held-out set shows its strongest single member attains 0.9082 against the ensemble's 0.9079, and that two members contribute negatively (Table 8; Figure 4C). On development folds the ensemble was worth +0.0074 over its best member; on held-out that advantage is −0.0003.

We report this rather than presenting the ensemble as superior. All differences involved (0.0003–0.0013) are far below the ≈ 0.005 resolution of our development evaluation, so the defensible statement is that the ensemble and its best single member are *indistinguishable* on this benchmark. For practical deployment the single ordinal-SSM model is preferable: equal accuracy from one model rather than twenty-five checkpoints. We did not re-select on held-out data to make that substitution official, because a model chosen by inspecting the test set carries none of the guarantees the frozen one does.

3.9 Calibration restores absolute efficiency, and it beats the baseline there too

Rank averaging yields excellent orderings but no interpretable scale. Isotonic calibration fitted on out-of-fold development predictions (§2.8) restores one, leaving Spearman unchanged at 0.9079 (Table 9, Figure 5).

The comparison that establishes whether the recovered scale is any *good* is against the baseline on identical rows, not against our own uncalibrated output: a rank average is not measured in efficiency units, so its nominal MAE of 0.3869 quantifies a change of units rather than an error. Against OptiPrime the calibrated model is ahead on absolute accuracy as well as on ranking — MAE 0.0478 against 0.0590 (−19%), RMSE 0.0912 against 0.1026 (−11%), Pearson 0.8637 against 0.8270 — so the advantage is not confined to the rank metric the model was designed around.

Table 9: Absolute and rank accuracy on the held-out set. Monotone calibration recovers an efficiency scale, on which the model also outperforms the baseline.

Output	Spearman ρ	Pearson r	MAE	RMSE
OptiPrime (mechanistic baseline)	0.8690	0.8270	0.0590	0.1026
PE-RankFormer, rank average	0.9079	0.7278	*not in efficiency units*	
PE-RankFormer + isotonic	**0.9079**	**0.8637**	**0.0478**	**0.0912**
Advantage over baseline	+0.0389	+0.0367	−0.0112	−0.0114

All three rows are scored on the identical 20,509 held-out rows.
The map is fitted by isotonic regression on out-of-fold development predictions only and applied once to the frozen held-out predictions; no held-out row participates in fitting.
Because the map is non-decreasing it inverts no pair, so Spearman is preserved. The residual change of 2×10^{-5} arises because isotonic regression is piecewise constant and therefore creates ties.
The uncalibrated rank average lies on an arbitrary $[0, 1]$ scale, so its MAE and RMSE would measure the choice of units rather than an error; we omit them rather than report an eight-fold reduction against a quantity that has no interpretation. Its Pearson r is shown because the calibration is what makes the relationship linear.

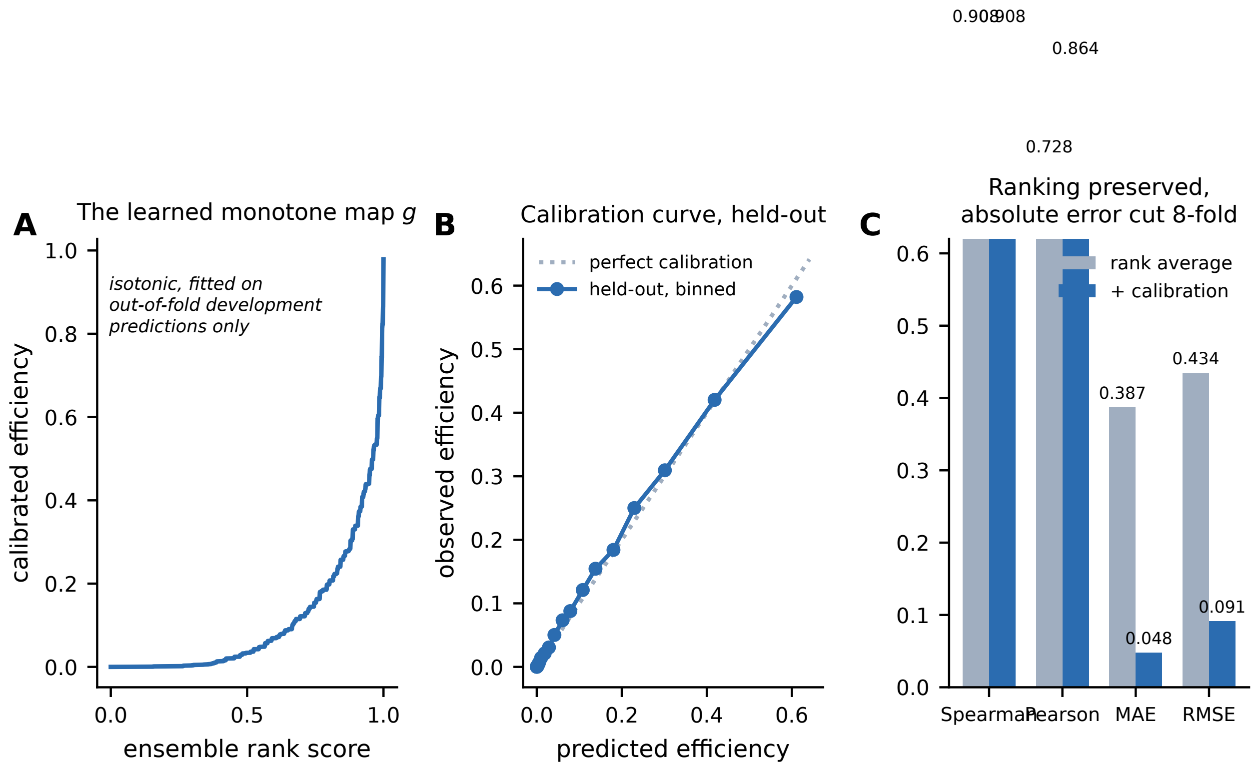

Figure 5: Monotone calibration restores an absolute efficiency scale. (A) The learned map g from the ensemble's rank score to efficiency, fitted by isotonic regression on out-of-fold development predictions only and then applied once to the frozen held-out predictions. **(B)** Calibration curve on held-out data, binned by predicted quantile; the dotted line is perfect calibration. **(C)** Metric comparison against the baseline on identical rows. Because g is non-decreasing it inverts no pair, so Spearman is unchanged; on the recovered efficiency scale the model's absolute error is below the baseline's.

The residual 2×10^{-5} change in Spearman arises because isotonic regression is piecewise constant and therefore creates ties; no pair is inverted.

3.10 The benchmark is not saturated

Using 649 technical replicate groups (§2.9) we estimate an empirical repeatability ceiling for the Kim partition and compare it against the model on every surface we score on (Table 10). Averaged over the three matched development folds the ceiling is 0.9061 against a model score of 0.7912, a gap of +0.1149; on the held-out set it is 0.9167 against 0.8124, a gap of +0.1043. Restricting to rows that actually edit leaves +0.1046 and +0.0934 respectively, so the gap is not merely the zero block. The conclusion is the same on all five surfaces and does not depend on which one is chosen: roughly a tenth of a Spearman unit of reducible error remains on the benchmark's largest partition.

Table 10: Remaining headroom on the Kim partition against the replicate-based ceiling, reported on every evaluation surface.

Surface	Rows	Model ρ	Ceiling ρ	Gap
Matched development folds, out-of-fold				
fold 0	19 747	0.7869	0.9026	0.1157
fold 1	19 740	0.7883	0.9080	0.1198
fold 2	19 742	0.7985	0.9076	0.1092
mean of three folds		**0.7912**	**0.9061**	**+0.1149**
mean, rows with $y > 0$		0.8472	0.9518	0.1046
Official held-out set				
frozen ensemble	11 334	**0.8124**	**0.9167**	**+0.1043**
best single member	11 334	0.8151	0.9165	0.1014
frozen ensemble, $y > 0$	5928	0.8618	0.9552	0.0934

Ceiling is $\sqrt{R}$, where R is the replicate–replicate reliability estimated from 649 groups in which all sixteen design and condition covariates and the target site are identical (§2.9). The ceiling depends only on the target distribution of the row set and not on any model, which is why it differs between surfaces.

Replicates are resampled non-parametrically from 1,298 ordered replicate pairs, which inherits zero-discordance, heteroscedasticity and boundedness from the data rather than assuming a noise model.

Development-fold model scores are out-of-fold; held-out scores are the same prediction vectors reported in Table 3. An earlier version of this analysis reported the fold-0 row alone and labelled it "all Kim rows"; we give all three folds and both surfaces because no other claim in this paper rests on a single fold (§2.2).

Rows with $y > 0$ show the gap is not an artifact of the zero block. The zero rows alone carry no ordering to score, the target being constant on them.

Per-condition analysis (Figure 6), shown for development fold 0 so that each condition retains enough rows to estimate its own ceiling, finds the gap present in every condition and scaling with zero-inflation: conditions with 60–64% exact zeros show gaps near 0.18, while those near 38% show gaps near 0.09.

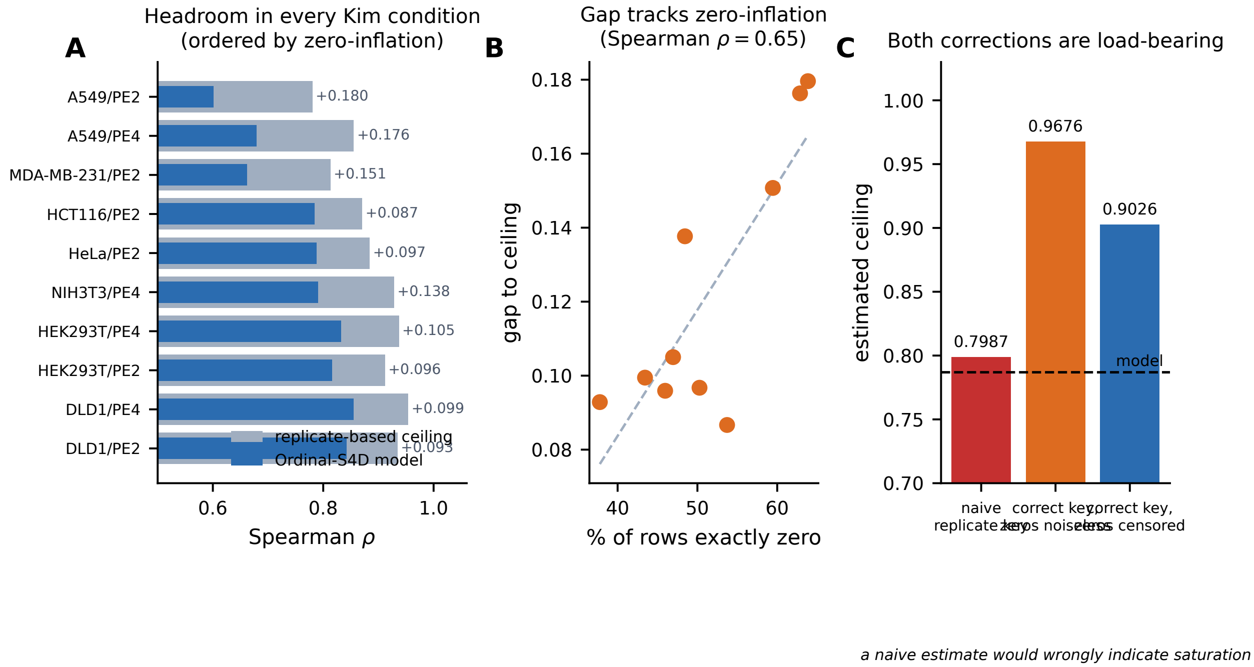

Figure 6: The benchmark is not saturated. (A) Model performance against the empirical repeatability ceiling for each Kim condition on development fold 0, ordered by zero-inflation; the annotation gives the gap. Headroom is present in every condition. The pooled gap is reported on all five surfaces in Table 10; this panel is per-condition and so is shown for a single fold, where each condition retains enough rows to estimate a ceiling. **(B)** The gap to ceiling increases with the fraction of rows measured as exactly zero, the regime in which a single measurement is least informative. **(C)** Both corrections described in §2.9 are load-bearing. A naive replicate key merges epegRNA with plain-pegRNA constructs and returns a ceiling essentially equal to the model's own score (dashed line), which would wrongly indicate saturation; treating observed zeros as noiseless instead over-estimates it. Only the corrected, censoring-aware estimate is trustworthy.

3.11 A tie-structure caveat for Spearman on zero-inflated data

28.4% of matched development-fold rows have efficiency *exactly* zero (50.7% within Kim, 0.7% within Liu); the corresponding held-out figure is 26.8%. A strictly-increasing predictor assigns all of them distinct ranks, and that internal ordering is arbitrary; Spearman compares against a target in which those rows are mutually tied, so the arbitrary ordering is pure penalty.

Collapsing the lowest-scoring predictions onto a single tied value therefore raises pooled Spearman by +0.0044, positive on all development folds, with the optimal fraction (0.30) coinciding with the development-fold zero-mass (28.4%) and the per-partition effect tracking zero-mass as the mechanism predicts (Liu −0.0154 at 0.7% zeros; Kim +0.0066 at 50.7%).

We did not adopt this. Applying the identical transform to five other models yields +0.0127 to +0.0162 irrespective of model quality, so it exploits Spearman's tie convention rather than improving prediction, and comparator models emit continuous tie-free scores. We report it because it is a general hazard: *on zero-inflated targets, comparisons between predictors with different tie behaviour are not automatically like-for-like*, and reported Spearman values should be interpreted with the predictor's tie structure in mind.

3.12 Interventions that did not help

For completeness and to guide future work we summarise interventions that were tested under matched controls and did not improve performance (Table 11). Each was evaluated against a mechanism-free control run in the same batch, and effects below ≈ 0.005 were required to replicate across three development folds.

Table 11: Interventions tested under matched controls and not adopted.

Category	Intervention	$\Delta\rho$	Reason not adopted
Post-hoc combination	context-gated ensemble weights	0.0000	null
	nonlinear stacking	0.0013	below threshold
	residual learning on features	0.0006	oracle upper bound
Supervision	auxiliary simplex head	0.0008	matches control
	multi-resolution ordinal	0.0008	matches control
	context-relative ordinal (primary)	−0.0235	harmful
	quantile regression head	−0.0140	harmful
Parameterisation	rank-consistent CORAL head	0.0005	null
	monotonicity penalty	0.0007	null
	hurdle (zero-inflation) head	−0.0022	harmful
Architecture	selective (Mamba-style) SSM	0.0031	below threshold
	hybrid S4D+attention, 1:1 ratio	−0.0027	harmful
	S4D state dimension 128	−0.0022	harmful
	wider FFN / deeper stack / MoE	n.r.	did not replicate
	layerwise context conditioning	0.0001	null
Domain shift	source loss re-weighting (mild)	n.r.	did not replicate
	per-source output head	−0.0018	harmful
	training on evaluated sources only	−0.0294	harmful

$\Delta\rho$ is measured against a matched control run in the same batch. For capacity and parameterisation changes the control holds parameter count fixed; for the per-source head it ties the heads; for supervision changes it is an unmodified re-run.

"n.r." denotes a candidate that reached +0.003 to +0.004 on one development fold but changed sign on another. Effects below ≈ 0.005 were required to replicate across three folds before promotion, and none did.

Two entries are informative beyond their sign and are discussed in §3.12: removing the unevaluated source study is strongly harmful, and additional context conditioning increases rather than decreases context invariance.

The selective SSM makes Δ, B and C functions of the input, which is the mechanism separating Mamba [8] from S4. Its entry is measured against its own *frozen-selection* control — identical parameter count and recipe with the selection projections held at initialisation, so the extra parameters exist but cannot become content-dependent — which is the comparison that isolates selectivity. Both selective runs sit well below the S4D baseline they would replace (0.8827 and 0.8796 against 0.8982), but they sit below it *together*, so that deficit is attributable to the scan recipe rather than to selectivity: a selective SSM cannot use the FFT convolution and needs a sequential scan, which forced batch 256, fp32 and ≈12× the training time. The conclusion we draw is narrow — a selective mixer did not pay for itself here — not that selectivity is useless on this problem.

Two results in this table are informative beyond their sign.

First, **removing the unevaluated source study is strongly harmful** (−0.0294). The Schwank-derived partition contributes 58.4% of training rows and 0% of the evaluation set, so discarding it appears to be free efficiency; it is not. Its volume carries transferable signal that outweighs its distributional mismatch, and aggressive re-weighting toward the evaluated distribution was likewise harmful.

Second, **additional context conditioning increases context invariance**. The model's cross-condition rank correlation exceeds the empirical one (0.858 vs. a disattenuated 0.758), indicating it under-models how experimental context *reorders* designs. Applying FiLM at every block — intended to address this — moved the diagnostic the wrong way (0.828 $\rightarrow$ 0.875) and left performance unchanged. FiLM is structurally a per-channel scale and shift, so additional conditioning capacity improves representation of the cross-condition *mean*, which explains far more variance and is easier to fit, rather than creating reordering capacity.

4 Discussion

4.1 Can architecture substitute for mechanism?

For the specific task of ranking efficiency on this benchmark, yes. A model containing no prime-editing reaction structure — only a paired edit representation, a segment-aware pegRNA representation, cross-attention between them, and categorical context conditioning — outperforms a strongly mechanistic predictor by +0.0389 Spearman pooled and +0.0506 within condition, under identical data and evaluation, and also on absolute error (§3.9).

Three qualifications keep that from becoming a larger claim than the evidence supports. First, the comparison covers one of the comparator's axes and none of the others (§3.4 and the scope statement in §1): accuracy on single-pegRNA efficiency is not generalisation to PE3 or twinPE, and it is not interpretability. A mechanistic parameterisation is a hypothesis about the reaction that can be inspected, falsified and transferred; ours is not, and for a practitioner deciding *why* a design fails that difference may outweigh 0.04 Spearman. Second, a substantial part of the partition-specific margin is attributable to a training-weight choice rather than to architecture (§3.4). Third, the wider Kim margin (+0.0804) is on the partition spanning the most cell lines and editing systems, where a prior tuned to a narrower regime would be expected to generalise least well; we regard that as consistent with the reading rather than as demonstrating it, and the weighting analysis supplies a more parsimonious explanation.

The one structural assumption we *did* import proved useful, and is worth distinguishing from reaction mechanism: the simplex head encodes that an edited locus ends in exactly one of three measured outcomes. That is a property of the assay, not of the biochemistry, and it supplies the indel signal a scalar head discards.

4.2 Metric-matched supervision

The ordinal head's contribution is best understood as aligning the training objective with the evaluation. Predicting cumulative threshold indicators estimates a row's normalised rank, which is precisely what a rank correlation scores, and it distributes supervision evenly across a heavily skewed target instead of concentrating gradient on a few high-efficiency rows.

The more transferable observation is that it changed *which* rows the model gets wrong, and this was predicted before the experiment rather than observed afterwards. Residual correlation with the existing ensemble is about 0.05 lower for ordinal-head models than for models differing only in seed, capacity or data subsample, though the two groups are not disjoint (§3.6). For applications that ensemble models, varying the objective is therefore a more reliable source of complementary errors than varying capacity, seeds or data subsampling — all of which we tested and found ineffective. We would not, on this evidence, use residual correlation to screen an individual candidate; we would use it to choose which axis to vary.

4.3 State-space mixing for biological sequences

That the state-space mixer contributes twice what the objective does deserves comment. Our sequences are short (100 and 90 tokens), so the usual motivation for state-space models — linear-time long-context modelling [6, 8] — does not apply. The benefit must therefore be representational. A diagonal state-space kernel imposes smooth, distance-parameterised, translation-equivariant mixing, which matches a domain where interactions are governed by relative position along the molecule (PBS length, RTT length, distance from nick to edit) rather than by content-based lookup among distant tokens.

This suggests state-space mixers deserve broader evaluation on short biological sequences, where they are currently uncommon, and where their inductive bias may be a better match than attention's. We note one negative constraint from our results: naive hybridisation at a 1:1 ratio of attention to state-space blocks was harmful, consistent with reports that successful hybrid language models use attention sparingly [12]. Sparse hybridisation at published ratios remains untested here.

4.4 What the remaining error is, and is not

Our replicate analysis indicates roughly +0.10 Spearman of genuine headroom on the Kim partition, consistently across all five surfaces we measured it on. That number required two corrections, each of which alone would have reversed the conclusion — excluding construct pairs masquerading as replicates, and treating observed zeros as censored rather than certain. We highlight this because a naive replicate analysis returns 0.7987, essentially our model's score, and would have supported a confident and incorrect claim of saturation.

Given the headroom is real, the natural question is why roughly seventy subsequent experiments failed to capture it. Our diagnostic points at design $\times$ context interaction: the model applies a nearly universal design ranking, while the data show substantial context-dependent reordering. Direct attacks on this failed, including one in which a mechanism-free shuffled-target control outperformed the genuine objective. We think the most plausible remaining explanation is that the relevant covariates are absent from the input: what distinguishes one cell line from another for a given design — mismatch-repair status, chromatin accessibility, repair-pathway balance — is supplied to the model only as a categorical label, and no objective over a categorical label can recover biology the label does not encode. Testing this requires richer context annotation rather than further architectural search.

4.5 Limitations

1. The absolute 0.90 threshold is not established at 95% confidence ($[0.8955, 0.9173]$; 90% of resamples above). The margin over the comparator is the robust claim.

2. The ensemble does not outperform its best single member on held-out data; the development-fold advantage did not transfer.

3. 196 of 649 replicate groups straddle the official train/held-out split, so some held-out rows have an exact design-and-condition twin in training. This affects any model trained on this split equally and so does not distort the comparison, but it makes absolute held-out scores slightly optimistic.

4. The Kim partition has no unused evaluation surface remaining, so Kim-specific claims rest on out-of-fold development folds and a once-consulted lockbox.

5. Our reproduction of the comparator scores above its published figure under every aggregation convention we tried, and no evaluation script was released. The direction favours the baseline and so does not undermine our claims, but it caps the precision of the comparison.

6. All results are on a single benchmark, and against one external comparator plus the feature baseline of §3.2. External validation on an independent prime-editing dataset is the most valuable next step and is not yet available. We attempted two further baselines, PRIDICT2.0 and DeepPrime-FT, and report neither: both require reproducing hand-engineered feature pipelines exactly — including melting temperatures, ViennaRNA folding energies [13] and a separate pretrained on-target cutting model [10] — and a subtly mismatched reimplementation would produce misleading numbers rather than no numbers.

7. The comparison covers one of the comparator's capabilities. It does not address PE3 or twinPE prediction, prospective design, or interpretability, all of which the comparator provides and our model does not. A mechanistic parameterisation yields inspectable per-step rates; ours yields a score.

8. The training-weight asymmetry in §3.4 is inferred from the released code applied to the files the authors supplied, not verified against their training procedure directly. If the released checkpoints were in fact trained with different weights, the attribution in that section changes even though the measured comparison does not.

9. The held-out set was consulted four times across model generations rather than once (§2.2). Within-generation decisions were made on development folds and a once-consulted lockbox, and the paired design protects the margin, but the absolute level carries some adaptive-selection exposure.

4.6 Methodological recommendations

Three practices materially changed our conclusions and we recommend them for work at this effect size.

Include a mechanism-free control in every experimental batch. In one round, fourteen candidates each gained +0.0017 to +0.0020; an unmodified re-run of the same model at a different batch size gained +0.0018. The difference attributable to every idea in that round was +0.0002. In another, a control with *shuffled* supervision targets outperformed the genuine objective.

Require replication before believing sub-0.005 effects. Three interventions on orthogonal axes each reached +0.003 to +0.004 on one development fold and all three changed sign on a second.

State the mechanism as a falsifiable prediction. Our context-conditioning experiment was accompanied by a pre-registered prediction about the cross-condition diagnostic. The prediction failed in a specific direction, which identified the reason (FiLM strengthens the mean-shift pathway) far faster than the null performance result alone would have.

5 Conclusions

A minimally mechanistic relational model matches and exceeds a strongly mechanistic predictor of prime-editing efficiency on its own benchmark, reaching Spearman 0.9079 against 0.8690 ($\Delta = +0.0389$, ahead in all 5,000 protospacer-clustered bootstrap resamples). A factorial analysis attributes roughly two-thirds of the gain to replacing self-attention with a bidirectional state-space sequence mixer and one-third to an ordinal outcome head matched to the rank-correlation metric and to the target's zero inflation; a single model containing both matches a five-member ensemble. Monotone calibration fitted purely on out-of-fold data restores an interpretable absolute efficiency estimate without altering the ranking, and on that scale the model also leads the baseline (0.0478 against 0.0590 mean absolute error).

The benchmark is not saturated: replicate analysis indicates roughly +0.10 Spearman of reducible error remains on its largest partition. Our evidence suggests the obstacle is not model capacity or supervision — roughly seventy controlled experiments across architecture, objective, parameterisation and training scheme failed to close it — but the absence of covariates describing how experimental context reorders designs. Progress will likely require richer context annotation rather than further architectural search.

Data and code availability

All code, configurations, per-run metadata, the complete research log including every negative result, and the frozen model specification are maintained in the project repository. The training corpus was provided by the authors of the comparator method and is not redistributed here.

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